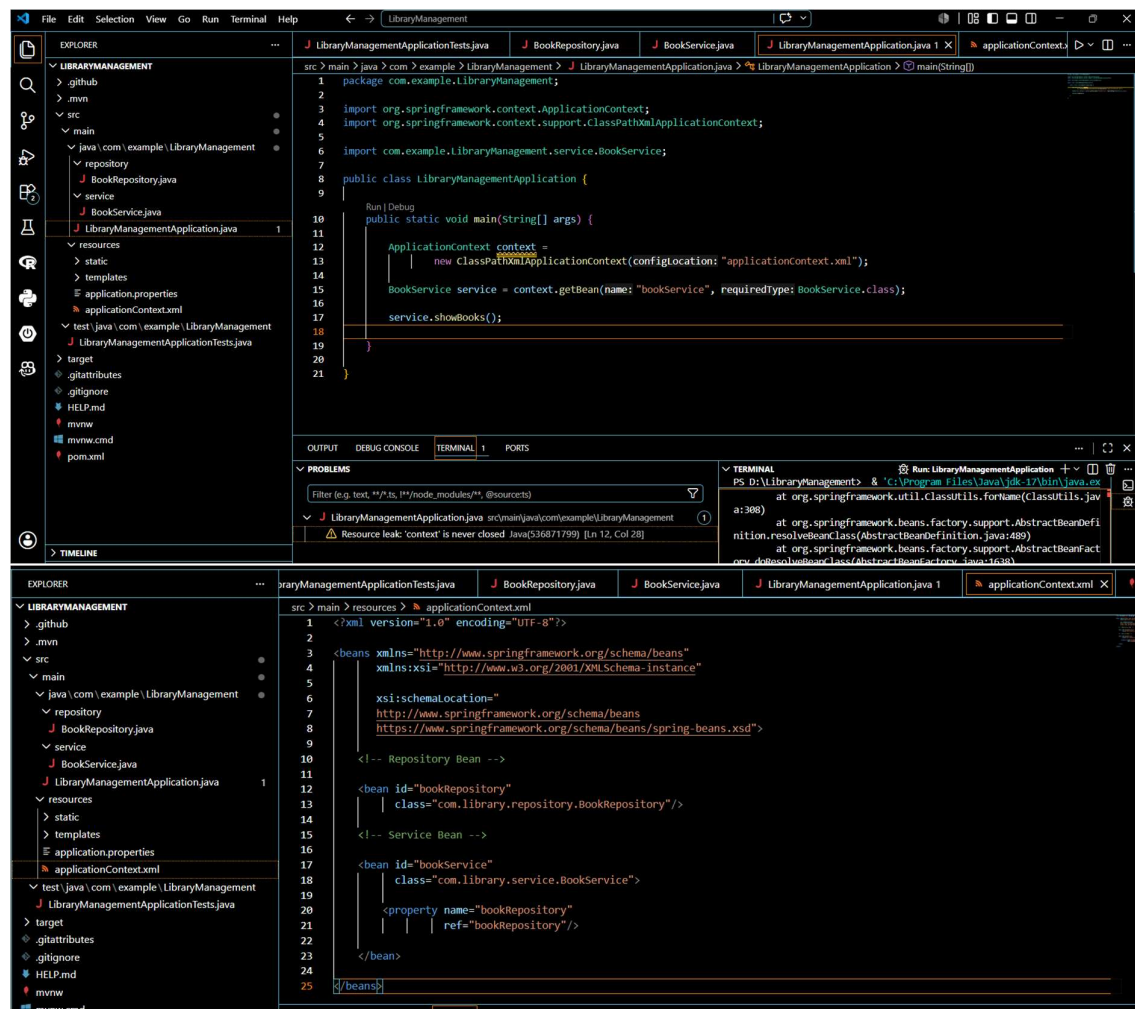


Exercise 1: Configuring a Basic Spring Application



File Edit Selection View Go Run Terminal Help

LibraryManagement

EXPLORER

- LIBRARYMANAGEMENT
 - .github
 - .mvn
 - src
 - main
 - java \ com \ example \ LibraryManagement
 - repository
 - BookRepository.java
- service
 - BookService.java
- LibraryManagementApplication.java 1
- resources
 - static
 - templates
- application.properties
- applicationContext.xml
- test \ java \ com \ example \ LibraryManagement
 - LibraryManagementApplicationTests.java

src > main > java > com > example > LibraryManagement > repository > BookRepository.java > {} com.example.LibraryManagement.repository

```
1 package com.example.LibraryManagement.repository;
2
3 public class BookRepository {
4
5     public void displayBooks() {
6         System.out.println(x: "Fetching books from repository...");
7     }
8
9 }
```

LibraryManagementApplicationTests.java BookRepository.java BookService.java LibraryManagementApplication.java 1 applicationContext.xml

src > main > java > com > example > LibraryManagement > LibraryManagementApplication.java > LibraryManagementApplication > main(String[])

```
1 package com.example.LibraryManagement;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.example.LibraryManagement.service.BookService;
7
8 public class LibraryManagementApplication {
9
10 }
```

OUTPUT DEBUG CONSOLE TERMINAL 1 PORTS

PROBLEMS

Filter (e.g. text, **/*.ts, !**/...)

- LibraryManagementApplicationTests.java

Java(536871799) [Ln 12, Col 1]

TERMINAL

Run: LibraryManagementApplication

```
PS D:\LibraryManagement> d:; cd 'd:\LibraryManagement'; & 'C:\Program Files\Java\jdk-17\bin\java.exe' '@C:\Users\chav
a\AppData\Local\Temp\cp_bauu7atw00d4ze2astaroovs9.argfile' 'com.example.LibraryManagement.LibraryManagementApplication
'
at org.springframework.context.support.AbstractApplicationContext.finishBeanFactoryInitialization(AbstractApp
licationContext.java:994)
at org.springframework.context.support.AbstractApplicationContext.refresh(AbstractApplicationContext.java:621)
at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:144)
at org.springframework.context.support.ClassPathXmlApplicationContext.<init>(ClassPathXmlApplicationContext.java:85)
at com.example.LibraryManagement.LibraryManagementApplication.main(LibraryManagementApplication.java:13)
Caused by: java.lang.ClassNotFoundException: com.library.repository.BookRepository
at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
at java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
at java.base/java.lang.ClassLoader.loadClass(ClassLoader.java:525)
at java.base/java.lang.Class.forName0(Native Method)
at java.base/java.lang.Class.forName(Class.java:467)
at org.springframework.util.ClassUtils.forName(ClassUtils.java:308)
at org.springframework.beans.factory.support.AbstractBeanDefinition.resolveBeanClass(AbstractBeanDefinition.java:489)
at org.springframework.beans.factory.support.AbstractBeanFactory.doResolveBeanClass(AbstractBeanFactory.java:1638)
at org.springframework.beans.factory.support.AbstractBeanFactory.resolveBeanClass(AbstractBeanFactory.java:1564)
... 12 more
C:\PS D:\LibraryManagement>
```

Exercise 2: Implementing Dependency Injection

```
src > main > java > com > example > LibraryManagement > repository > BookRepository.java > BookRepository
1 package com.example.LibraryManagement.repository;
2
3 public class BookRepository {
4
5     public void displayBooks() {
6         System.out.println(x: "Fetching books from repository...");
7     }
8 }

LibraryManagementApplicationTests.java | BookRepository.java | BookService.java X | LibraryManagementApplication.java 1 | applica

src > main > java > com > example > LibraryManagement > service > BookService.java > BookService
1 package com.example.LibraryManagement.service;
2
3 import com.example.LibraryManagement.repository.BookRepository;
4
5 public class BookService {
6
7     private BookRepository bookRepository;
8
9     // Setter Injection
10    public void setBookRepository(BookRepository bookRepository) {
11        | this.bookRepository = bookRepository;
12    }
13
14    public void showBooks() {
15        | System.out.println(x: "Book Service Called");
16        | bookRepository.displayBooks();
17    }
18 }

LibraryManagementApplicationTests.java | BookRepository.java | BookService.java | LibraryManagementApplication.java 1 X | applicationContext.xml

src > main > java > com > example > LibraryManagement > LibraryManagementApplication.java > LibraryManagementApplication
1 package com.example.LibraryManagement;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.example.LibraryManagement.service.BookService;
7
8 public class LibraryManagementApplication {
9
10    Run | Debug
11    public static void main(String[] args) {
12        | ApplicationContext context =
13        | | new ClassPathXmlApplicationContext(configLocation: "applicationContext.xml");
14        |
15        | BookService service =
16        | | context.getBean(name: "bookService", requiredType: BookService.class);
17        |
18        | service.showBooks();
19    }
20 }

PS D:\LibraryManagement> d:; cd 'd:\LibraryManagement'; & 'C:\Program Files\Java\jdk-17\bin\java.exe' '@C:\Users\cha
a\AppData\Local\Temp\cp_bauu7atwo0o4ze2astaroovs9.argfile' 'com.example.LibraryManagement.LibraryManagementApplicatio
.
Book Service Called
Fetching books from repository...
PS D:\LibraryManagement>
```

```
src > main > resources > applicationContext.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2
3  <beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5
6      xsi:schemaLocation="
7          http://www.springframework.org/schema/beans
8          https://www.springframework.org/schema/beans/spring-beans.xsd">
9
10     <!-- Repository Bean -->
11     <bean id="bookRepository"
12         class="com.example.LibraryManagement.repository.BookRepository"/>
13
14     <!-- Service Bean -->
15     <bean id="bookService"
16         class="com.example.LibraryManagement.service.BookService">
17
18         <property name="bookRepository"
19             ref="bookRepository"/>
20     </bean>
21
22 </beans>
23
```

Exercise 4: Creating and Configuring a Maven Project

```
pom.xml X
pom.xml
2  <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
32  <dependencies>
36      </dependency>
37
38      <dependency>
39          <groupId>org.springframework.boot</groupId>
40          <artifactId>spring-boot-starter-test</artifactId>
41          <scope>test</scope>
42      </dependency>
43  </dependencies>
44
45  <!-- Spring Context -->
46  <dependency>
47      <groupId>org.springframework</groupId>
48      <artifactId>spring-context</artifactId>
49  </dependency>
50
51  <!-- Spring AOP -->
52  <dependency>
53      <groupId>org.springframework</groupId>
54      <artifactId>spring-aop</artifactId>
55  </dependency>
56
57  <!-- Spring Web MVC -->
58  <dependency>
59      <groupId>org.springframework</groupId>
60      <artifactId>spring-webmvc</artifactId>
61  </dependency>
62
63 </dependencies>
64 </dependencies>
```

```
Run: LibraryManagementApplication
PS C:\Users\chava\Downloads\LibraryManagement (1)\LibraryManagement> mvn clean
compile
7 KB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.5.0/plexus-utils-3.5.0.jar (267 KB at 1.2 MB/s)
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug release 17] to target\classes
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 11.564 s
[INFO] Finished at: 2026-06-27T19:32:40+05:30
[INFO]
```